ARTD 326 - Fall 2025

The Mycelium Dragon

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Our Process

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Project Concept & Rationale

In eastern cultures dragons are seen as symbols of *power, wisdom, good fortune, and natural harmony.*

The Sibel Center of Design (SCD) is very popular on campus, especially during exam season, when the stakes are high and stress is higher.

Our dragon helps remind students that they are awesome and capable. We hope to create a boost in the morale of students with our installation.

Our location looks over one of the most popular study locations, acting as a **beacon of courage and hope.**



Planning

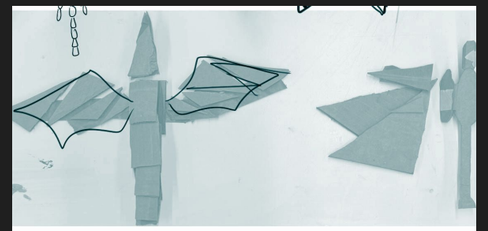
- We chose the area near **classroom 60** in the basement.
- The dragon would hang from the **bottom of the stair set** on the first level.
- Developed a rough plan for how we wanted to **grow our mycelium** and how we would **construct the final dragon.**

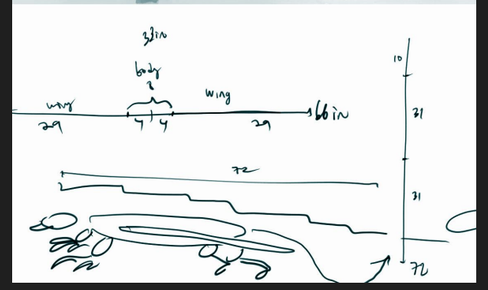
Design

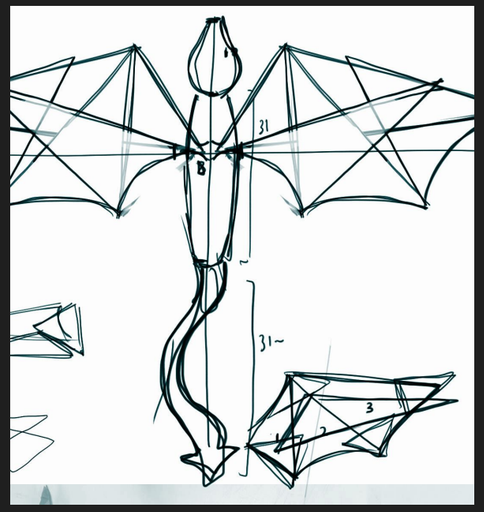
Spent a lot of time figuring out how we were going to approach our dragon. We had to be **realistic** and broke everything down to **simple shapes** to help us plan/make growing easier.

Our main focuses were:

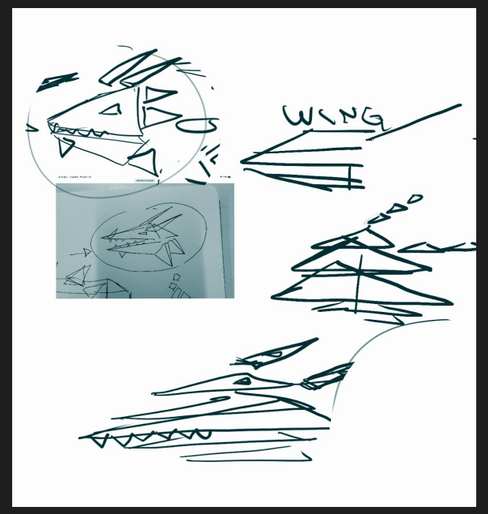
- figuring out the scale
- making sure things were proportional.
- gain an understanding of how much mycelium we were going to need.



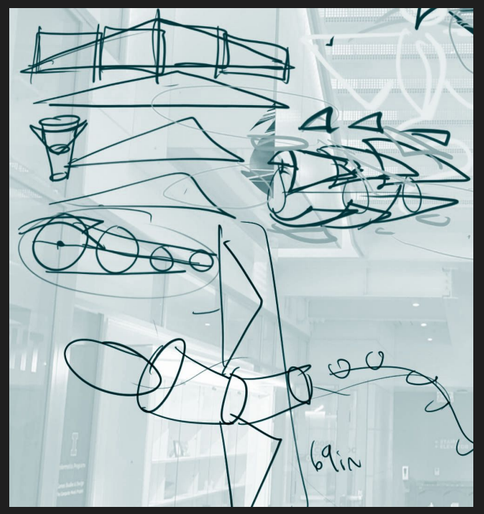


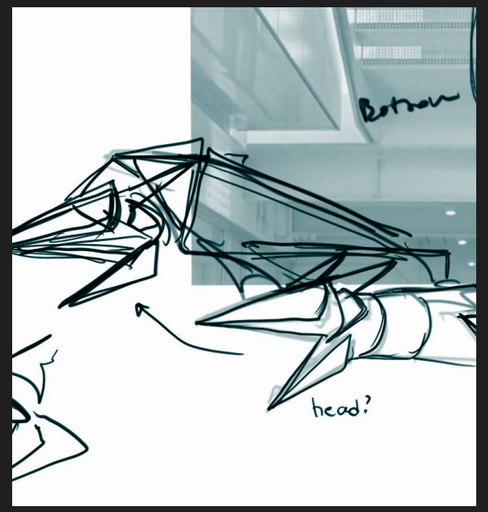


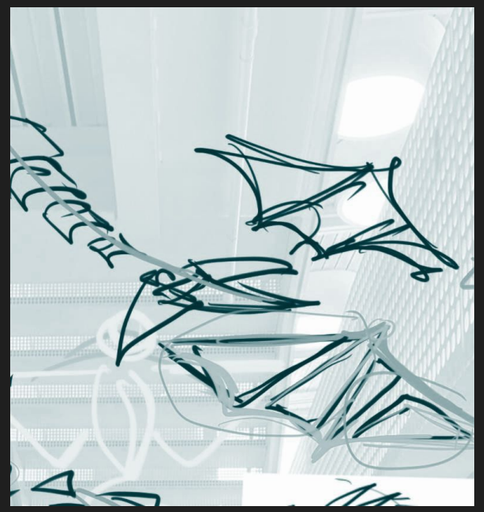


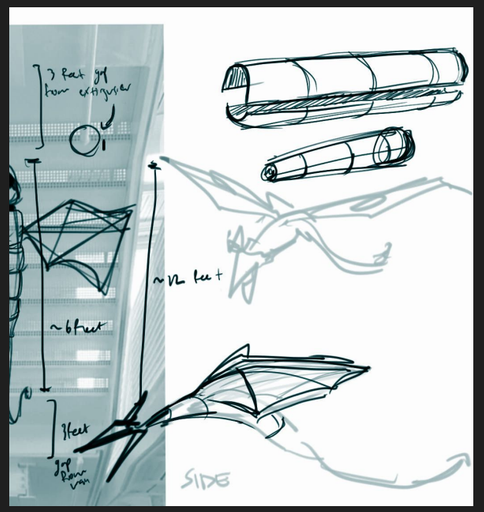




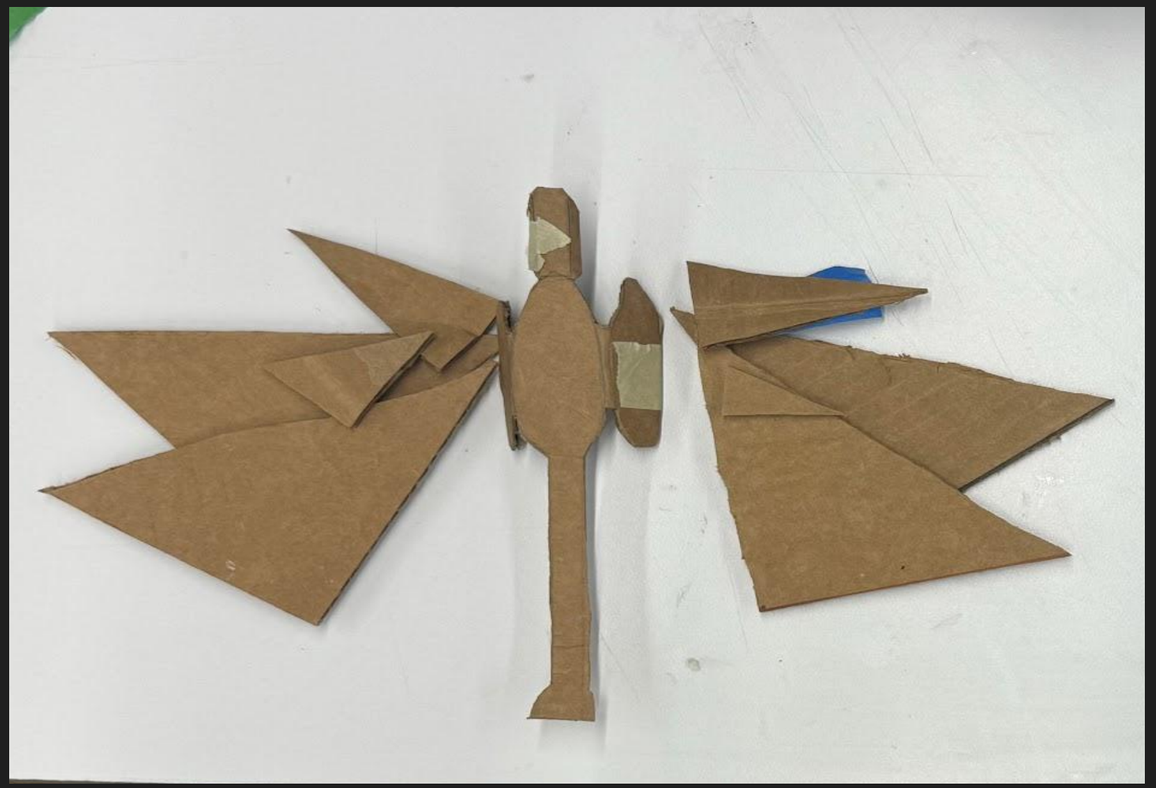






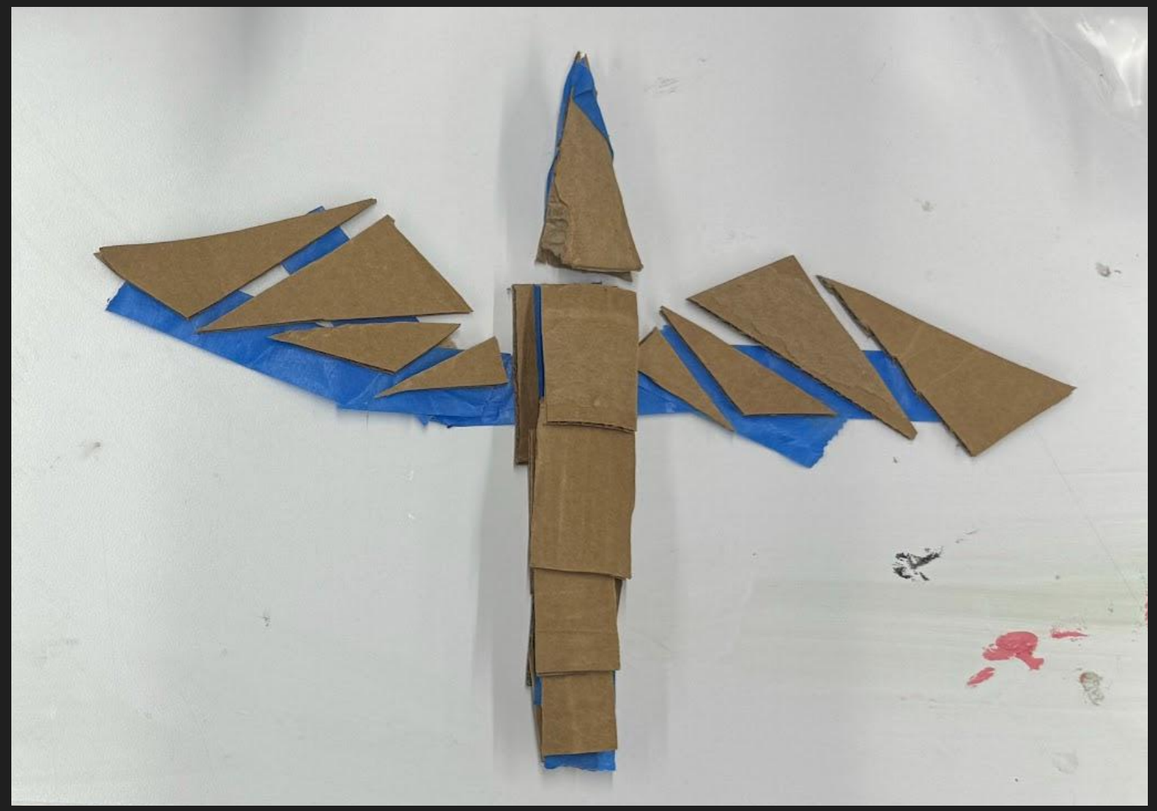


Prototypes



First Design

Very geometric, small number of pieces but they have to be larger and they would be layered to create a 3D effect.



Second Design

Smaller pieces that form scales and wings held together by some base/frame. Smaller pieces but in greater quantity.

Growing Plan









Material Gathering

Buy necessary supplies. Use office paper and package cardboard waste for sustainability.

Sanitation

Clean off surfaces, paper, cardboard stoppers, and growing trays.

Sheet Formation

Sanitized paper, layer of blended mycelium/sawdust mixture, Top paper layer.

One sandwich per tray; trays separated by cardboard stoppers.

Packing

Placed layered trays in air-filtered bags, sealed with tape. Prevented mold and regulated moisture.

Storage

Bags and trays placed in a dark locker in a dry basement.

Growing Challenges

Time

Some took too long to grow; others grew too quickly and started fruiting..

We grew so many batches that we lost track of when we started growing.

Mold

We couldn't tell what was mold or was mycelium. Contamination was hard to monitor.

Moisture

After switching from paper towel to paper and growing location a lot of sheets had a lot of moisture.

Airflow

The way we stacked our sheets led the top sheets to receive the most oxygen, resulting in inconsistent sheets.

The orientation of the sheets also sometimes led to deformed sheets.

Scale

Struggle between large sheets or more small sheets was difficult until we understood the final design's scale.

Temperature / Humidity

We had to switch growing locations at the end due to the amount of mycelium we were growing.

New areas were more humid and colder which impacted our sheets.

Material

We spent a lot of time figuring out how we would make the frame or whether there would even be a frame.

In the end we were able to overcome most of our challenges by just growing a lot of mycelium. This gave us a lot of freedom, choice, and flexibility when it came to manufacturing.

Growing Results

Total # of sheets grown:

40+ sheets

Our growing process was streamlined so we could grow a lot with low effort, allowing us to focus on other aspects like the frame and presentation.

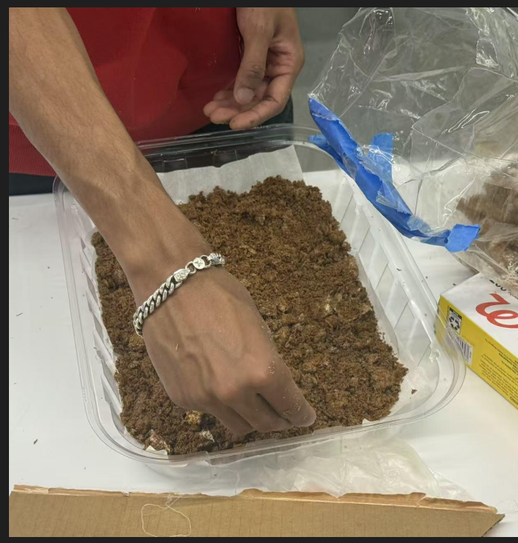
Total # of sheets used:

26 sheets





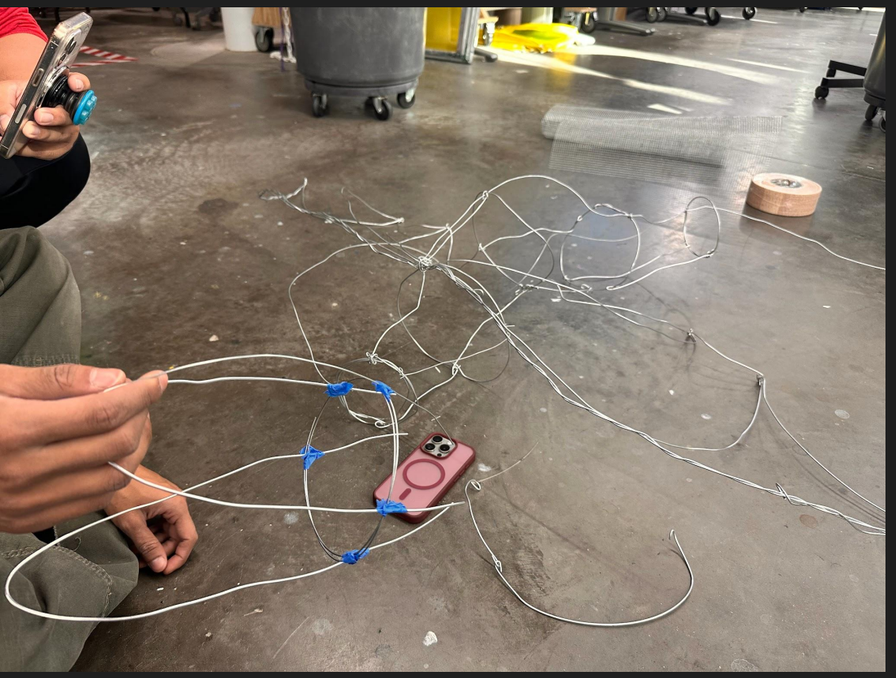








Frame Construction





5-6 Days

Total construction time.

Materials:

- 2 rolls of drywall tape
- 100 ft. 18 gauge hanging wire
- Pliers/wire cutters
- Scissors

Steel wire was bent to create a basic “skeletal” system of the dragon.

Drywall tape acted like skin, creating a base for the mycelium to stick to.

When completed, the frame was almost 15-20 lbs.

Heat Treatment





- To increase the durability of our mycelium sheets, we utilized heat pressing and heat guns to dry them out.
- Increased moisture in some sheets required multiple rounds of heat treatment.

Finished Frame + Mycelium



Final Frame

Finished frame before adding horns and scales



Glue

Mixture of spray adhesive for large areas and hot glue for touch ups and detailed areas

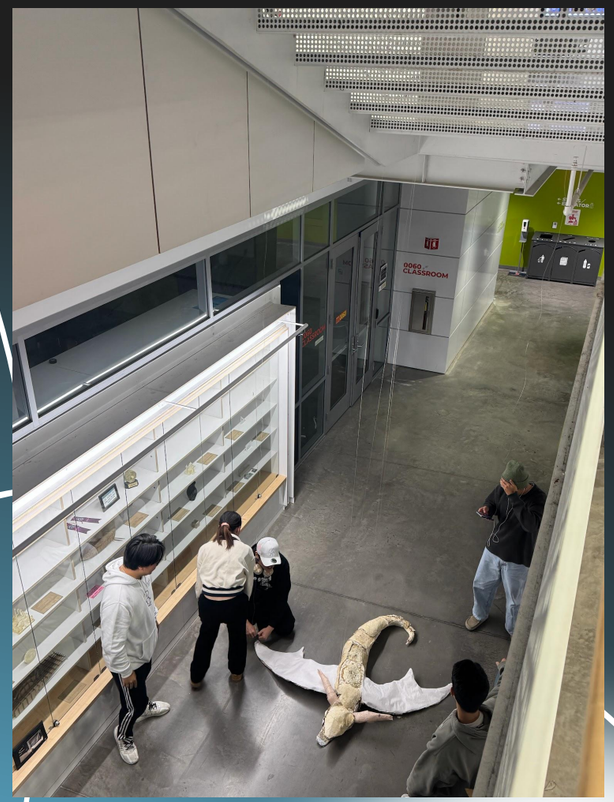


Finishing touches

Fixed small patches with hot glue and cut mycelium to fit. Chose every panel deliberately for color and patterns.

Used paper for top of wings and left some tape visible for weight and aesthetics.

Installation





Transport & Preparation. We used a cart to move the dragon, and let down the lines of fishing wire from the stair set.

Pulling it up. Tied string to the head, body, tail, and each wing with the wire going back up. One person stayed down to give direction and everyone else manned a string

These hooks played an important role in helping us pull the dragon up

Keeping it in place. We tied and taped down the fishing wire at the top of the stairs.

Exhibition

- Our dragon was only on display for almost 2 weeks at SCD.
- We presented it to the class along with this poster.
- The poster gives a brief overview of our process.







Thank You!

ARTD 326
Sustainability & Manufacturing
Fall 2025